

ETHAN YOUNGIN SHIN

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RESEARCH INTERESTS

Turbulence, Boundary-Layer Meteorology, Air-Sea Interactions, Uncertainty Quantification, Inverse Problems, Environment-Energy System Interactions, Engineering for Global Development

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA, United States

Sep. 2022 – Present

Ph.D. Candidate in Civil and Environmental Engineering

Thesis Advisor: Michael F. Howland

Yonsei University, Seoul, Republic of Korea

Mar. 2016 - Aug. 2022

B.S. in Mechanical Engineering

Honors: *magna cum laude*

RESEARCH EXPERIENCE

Howland Lab (Prof. Michael Howland), Massachusetts Institute of Technology

Graduate Research Assistant

Sep. 2022 - Present

- Coupled multifidelity simulations of atmospheric boundary layer flows with forward and inverse uncertainty quantification methods, to quantify and reduce uncertainty in turbulence parameterizations across atmospheric conditions
- Investigated the sensitivity and dependence of wall-stress errors (log-layer mismatch) in wall-modeled large-eddy simulations of geophysical flows to numerical setup and surface parameters
- Developed a probabilistic (Bayesian) method to estimate surface roughness parameters for Monin-Obukhov similarity theory from high-fidelity simulations and field observations, producing estimates with quantified uncertainty

Innovative Design and Integrated Manufacturing Lab (Prof. Sung-Hoon Ahn), Seoul National University

Undergraduate Research Intern

Dec. 2020 - Feb. 2022

- Conducted field demonstration of a polling-based energy management system with an off-grid solar power plant in Tanzania
- Investigated feasibility of a smart electric mobility operating system for integration with off-grid solar power plants in Tanzania

Multiphysics Energy System Laboratory (Prof. Jongsup Hong), Yonsei University

Undergraduate Research Intern

Jul. 2020 - Aug. 2020

- Conducted a thermofluidic analysis of a solid oxide fuel cell (SOFC) cooling system using ANSYS Fluent
- Designed and tested the effects of rib turbulators on SOFC interconnect fuel channels

PUBLICATIONS

Journal Publications

3. **Shin, E. Y.**, Kale B., and Howland, M. F. Navigating parameter uncertainty and model-form error in the data-driven development of parameterizations in Earth system models. *In preparation.*
3. **Shin, E. Y.** and Howland, M. F. Probabilistic inference of surface parameters for Monin-Obukhov similarity theory. *In review.*
3. **Shin, E. Y.**, Kale B., and Howland, M. F. Connecting the forward problem to the inverse problem in uncertainty quantification of Earth system models using fast emulators. *In review.*
3. **Shin, E. Y.** and Howland, M. F. Accelerated Bayesian calibration and uncertainty quantification of RANS turbulence model parameters for stratified atmospheric boundary layer flows. *Boundary-Layer Meteorology*, 192(3).
2. **Shin, E. Y.**, Yang, X. I. A., and Howland, M. F. Addressing grid convergence and log-layer mismatch in wall modeled large eddy simulations of geophysical flows over rough surfaces and canopies. *Boundary-Layer Meteorology*, 191(42).
1. Rhee, H., Im, H., Manongi, F., **Shin, Y.**, Song, H., Jung, W., Ahn, S. (2021). Smart Electric Mobility Operating System Integrated with Off-Grid Solar Power Plants in Tanzania: Vision and Trial Run. *Journal of Appropriate Technology*, 7(2), 127-135.

Conference Proceedings

1. **Shin, E. Y.**, Chan, M., Wang, J., Zahtila, T., Gorle, C., Iaccarino, G., and Howland M. F. (2024). Multifidelity modeling and uncertainty quantification of heterogeneous roughness. *Proceedings of the Summer Program, Center for Turbulence Research, Stanford University*.

CONFERENCE PRESENTATIONS AND POSTERS

5. **Shin, E. Y.**, Kale B., and Howland, M. F. Reducing parametric uncertainty in earth system models with efficient emulators. *MIT Climate & Sustainability Consortium: Poster Session on AI for Sustainability, Apr. 13, 2026, Cambridge, MA, USA*.
5. **Shin, E. Y.**, Yang, X. I. A., and Howland, M. F. Addressing grid convergence and log-layer mismatch in wall modeled large eddy simulations over rough surfaces and canopies. *77th Annual Meeting of the APS Division of Fluid Dynamics, Nov. 23-25, 2025, Houston, TX, USA*.
4. **Shin, E. Y.** and Howland, M. F. Bayesian inversion to quantify parameter uncertainty in RANS turbulence models for stratified atmospheric boundary layer flows. *AMS 25th Symposium on Boundary Layers and Turbulence, Jun. 17-20, 2025, Turin, IT*.
3. **Shin, E. Y.** and Howland, M. F. Accelerated uncertainty quantification of RANS turbulence models for stratified atmospheric boundary layer flow. *AGU24 Annual Meeting, Dec. 9-13, 2024, Washington, DC, USA*.
2. **Shin, E. Y.**, Chan, M., Wang, J., Zahtila, T., Gorle, C., Iaccarino, G., and Howland M. F. Multi-fidelity modeling and uncertainty quantification of heterogeneous roughness. *77th Annual Meeting of the APS Division of Fluid Dynamics, Nov. 24-26, 2024, Salt Lake City, UT, USA. (Invited)*
1. **Shin, E. Y.** and Howland, M. F. Uncertainty quantification of RANS turbulence model parameters with ensemble Kalman methods and machine learning. *76th Annual Meeting of the APS Division of Fluid Dynamics, Nov. 19-21, 2023, Washington, DC, USA*.

AWARDS AND SCHOLARSHIPS

1st Place Best Student Oral Presentation, American Meteorological Society (25th Symposium on Boundary Layers and Turbulence, Jun. 17-20, 2025)
CEE Mathworks Fellowship, Massachusetts Institute of Technology (Fall 2023, Spring 2024)
Louis Berger Fellowship, Massachusetts Institute of Technology (Spring 2023)
Schoettler Fellowship, Massachusetts Institute of Technology (Fall 2022)
Highest Honors, Yonsei University (Spring 2021)
Honors, Yonsei University (Fall 2016, Spring 2019, Spring 2020, Fall 2020)
Merit-based Scholarship, Yonsei University (2019, 2020, 2021, 2022)
National Scholarship, Korea Student Aid Foundation (2016, 2019, 2020)
Lee Choong Kon Pay Forward Scholarship (Spring 2019)

TEACHING EXPERIENCE

Teaching Assistant

1.861/1.086 Physics and Engineering of Renewable Energy Systems (Prof. Michael F. Howland) Mar. 2026 - Present

- Delivered one lecture, created 4 problem sets and a midterm, and held two office hours per week.

Teaching Assistant

1.61/1.061 Transport Processes in the Environment (Prof. Heidi Nepf) Sep. 2025 - Dec. 2025

- Delivered two lectures, created a problem set, held two office hours per week, and designed a laboratory class on particle advection and diffusion.

Guest Lecturer

Leaders in Industry-University Cooperation 3.0 Jul. 2023 - Present

- Delivered guest lectures on modeling wind in the atmospheric boundary layer to undergraduate students from Korean universities (Jeonbuk National University, Sookmyung Women's University)

Participant

MIT Teaching & Learning Lab Feb. 2023 - Dec. 2023

- Completed the four tracks of the Grad Teaching Development Tracks (equivalent to Kaufman Teaching Certificate Program), a set of workshops intended for graduate students interested in improving their teaching skills

Part-time Teacher

ASAP Academy Jul. 2019 - Aug. 2020

- Taught high school math, physics, and English grammar to classes of up to 7 students

ACADEMIC SERVICE

Reviewer

Boundary-Layer Meteorology, Wind Energy

Sep. 2025 – Present

Student Member

Committee on Boundary Layers and Turbulence, Americal Meteorological Society

Feb. 2025 - Present

Reviewer

Tackling Climate Change with Machine Learning, NeurIPS 2025

Sep. 2025

VOLUNTEER EXPERIENCE

Volunteer

Loaves and Fishes Meal Program

Dec. 2022 - Sep. 2024

Volunteer

Yonsei Habitat for Humanity

Mar. 2020 - Jun. 2022

Student Volunteer

Incheon Support Center for Foreign Workers

Mar. 2016 - Jun. 2016

LEADERSHIP EXPERIENCE

Panel Director

MIT Energy Conference

Oct. 2023 - Mar. 2024

President

Charles River Running Crew, MIT KGSA

June. 2023 - June. 2024

15th Class President

Yonsei Habitat for Humanity

Mar. 2020 - Jun. 2021

Student Ambassador

Yonsei-Waseda Exchange Program

Jul. 2016, Feb. 2019

Police Bus Driver

Auxiliary Police, Republic of Korea (Military Service)

Apr. 2017 - Dec. 2018